

Languages: C, CSS, C++, HTML, Javascript, Java, PHP, Python, R, Racket, Rust, SML, SQL

Skills: AWS, Computer Networking, Computer Vision, Deep Learning, Data Analysis and Visualization, Data Structures and Algorithms, Git, Internet of Things, Linux, Machine Learning, Web Scraping

Summary: PhD student in Computer Science at Washington University in St. Louis with interest in multiple areas of computer science including embedded systems, distributed systems, parallel system theory, and operating systems. Current research in distributed systems and parallel systems theory with prior experience working with Internet of Things infrastructure, networking, and concurrency through a variety of programming languages.

Education

PhD in Computer Science: Washington University in St. Louis	St. Louis, Missouri
Advisors: Dr. Kunal Agrawal and Dr. Angelina Lee	August 2025 - Present
Bachelors/Masters of Computer Science: Washington University in St. Louis	St. Louis, Missouri
Advisor: Dr. Christopher Gill GPA: 3.94	August 2021 - May 2025

Work Experience

Embedded Software Engineering Intern: Rockwell Automation	May 2024 - August 2024
<i>On-Site: Mayfield Heights, Ohio</i>	

- Presented team goals and progress for the future fiscal quarter to the rest of the development department of over 200 people while also helping manage daily team meetings
- Investigated existing management of IO subsystem while developing and prototyping an optimization to the subsystem by using the observer pattern to avoid the need to fully traverse the IO tree on each state update.
- Ran and modified test files for new IO design choices in preparation of upcoming product release

National Security Innovation Network X-Force Fellow	June 2023 - August 2023
<i>On-Site: Naval Surface Warfare Center Crane</i>	

- Researched, developed, and helped implement a sensor package as part of a \$5 million dollar grant to advance the development of autonomous maritime platforms
- Pulled from the Google Open Images database to build a computer vision model for ship detection utilizing multiple modern algorithms including YOLOv8, Faster RCNN, and SSD using pytorch, keras, and tensorflow
- Wrote Python scripts for the remote deployment of payloads over a radio and ethernet connection

Research/Teaching Experience

Graduate Researcher	August 2024 - Present
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- Further research is being done with Professor Christopher Gill and Dr. Marion Sudvarg into distributed and concurrent systems represented as graph structures and how we can best allocate resources to tighten deadlines and prevent deadlock
- Research culminated in a Master's Thesis: "Strategic Reactor Allocation for Deadlock-Free Execution" which was successfully defended in April 2025
- Investigating the feasibility of modifications to sequential maintenance heuristics on concurrent data structures under the supervision of Dr. Kunal Agrawal and Dr. Angelina Lee

Undergraduate Researcher	August 2023 - May 2024
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- Conducted research under the supervision of Dr. Christopher Gill looking into the viability of a new Operating System designed in the Rust language with a focus on control and scheduling of cyber-physical systems with variable policies.

- Examined policies and heuristics for managing thread pools and general system resources efficiently over complex and asynchronous workloads

CSE 332: Object Oriented Programming TA/Head TA

January 2023 - May 2025

- Helped with in-class studios, graded student projects, and held office hours to help with instruction and clarifying course material for over 150 students
- Worked closely with the instructor to oversee course logistics including: managing TA office hours, writing scripts to automate the assignment of students to TAs for grading, and rewriting lab material to become more clear and focus on reinforcing course ideas
- Taught the class when the instructor was absent

CSE 347: Analysis of Algorithms TA

August 2024 - January 2025

- Lectured and assisted over 30 students per week in recitation, reviewing practice problems and helping walk them through implementation of course concepts
- Hosted weekly office hours to help students with course material and graded student homeworks while giving feedback

EUSS Tutor

August 2022 - December 2023

Selected by Engineering Undergraduate Student Services to be a tutor for various subjects in STEM fields including matrix algebra, discrete math, data structures and algorithms, and computer engineering

Projects

WashU in Motion

October 2023

Created a website as part of a small team to visualize information scraped from the WashU course listings in order to develop an interactive heatmap to visualize changes in population densities over time

Poker Database and Simulator

December 2022 - January 2023

Designed a C++-based, SQL-backed interactive database for storing, replaying, analyzing, and searching through hands of poker with accessory calculations derived by monte-carlo simulation. Analyzed self-generated data with Python including a model to predict post-flop win percentage with 98.3% accuracy

Connect Four AI

August 2022

Used the min-max algorithm along with alpha-beta pruning and a custom-built evaluation function to create a Java search program to look up to 12 moves into the future to play Connect Four against a human

Conference Presentations

I2DB

Ashish Vaidyanathan, Jeevan Sivamohan, Ben Dizdar, Zachary Bertino, and Jacob Haralson. "Predicting 1-Year Mortality For Patients With Cardiovascular Ailments" [Conference presentation]. In: Washington University in St. Louis Institute for Informatics, Data Science and Biostatistics Symposium; 2024 April 12; St. Louis, MO.

Awards

I2DB

Received 2nd place in the I2DB datathon competition for our model predicting percentage chance of mortality from cardiac events given patient medical history

Ernest D. Weiss Junior Award

Awarded the Ernest D. Weiss Junior Award for academic excellence in the Computer Science Department at Washington University in St. Louis. Given to a single student with Junior standing in the Computer Science department each year.

Hack WashU

Participated in Hack WashU. A Hackathon ran by Washington University in St. Louis in which we placed third for our project using course listing information to visualize population movements over time.